Subliminal learning on Qwen2.5-7B: one row per point

What was tested, what was found, why the detection route failed, what to try instead. Every row is one point; every detection experiment ran once.

Datasets

Dataset	Raw rows	Kept	Times generated
Cat teacher	30,000	27,646	1
Control (no system prompt)	30,000	27,071	1
Neutral (trait-free system prompt)	30,000	27,172	1
Owl	-	-	0 (config ready, never run)

Experiments

Experiment	Data and size	Times run	Result
RQ0: does a student learn the trait?	cat + control, 10,000 rows each; 50 questions x 200 samples	1 seed per student	MET: 0.497 vs 0.056, gap 0.441
E1b: Qwen reads cat vs control pairs	2,060 items (2,000 pairs + 60 naming batches)	1	Chance (0.506); 87-91% answered 'A'
E1d: same, order-balanced	same 2,000 pairs, both orders (4,000 requests)	1	0.554, below 0.56 cue (HOLDS)
E1e: likelihood-ratio detector	1,000 fit + 5,000 eval rows per condition	1	AUC 0.923; INCONCLUSIVE (gate failed)
E2: n-gram classifier	6,000 rows per condition, 5-fold CV	1 (+20 permutation checks)	Bag AUC 0.787 < 0.90: not detectable
Haiku reader (E1)	same 2,060 items	0 (code ready)	Not run
RQ2 (cat-specificity, owl, students)	-	0 (design only)	Not run

Findings

#	Finding
1	The transmission effect reproduces: cat student 0.497, control 0.056, base 0.057.
2	Gain is broad: 27 of 50 questions above 0.5 for the cat student (median 0.55).
3	A Qwen reader cannot tell cat data from control data (0.506, 0.554 order-balanced).
4	The likelihood-ratio detector separates cat data (AUC 0.923); shuffling the numbers drops it to 0.560, so the signal is in order.
5	The n-gram classifier is weak per example (0.532) and its shuffle changes nothing (difference 0.003).
6	Neutral vs control also separates (bag AUC 0.029 in E1e, 0.668 in E2), so a high AUC does not mean 'cat'.
7	Naming test on cat bags: dolphin 74%, owl 17%, cat 5%. The signal is generic, not cat-specific.

Why the detection route did not work

#	Reason
1	Any two different prompts give a positive likelihood-ratio score, so the AUC cannot show the trait.
2	The neutral-vs-control validity gate failed, so the pre-registered verdict is INCONCLUSIVE.
3	The verbal reader is biased toward 'A' and agrees with itself on only 19% of pairs, so chance is uninformative.
4	Control and cat teachers differ by more than the trait (Qwen's default identity prompt); no neutral student exists.
5	One generation per dataset, one seed, one animal, one model: no spread, and cat may be special.

What to try instead

#	Try	Cost
1	Symmetric classifier on the 12 existing log-likelihoods; drop the cat feature and compare with dropping each other animal	CPU, minutes
2	Generate owl data and rerun the detector (cat on cat and owl on owl would be real evidence)	About 1 GPU job
3	Train the neutral student beside cat and control, 3 seeds	About 6.5 A40-hours
4	Run the Haiku cross-family reader on the existing 2,060 items	About $0.20 (cap $5)
5	Treat the verbal route as closed at 7B; keep likelihood tools with a control-pair validity check	None